

IFC for Smartie-pants



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Bimforce



UNIVERSITAT POLITÈCNICA DE VALÈNCIA

VALENCIA

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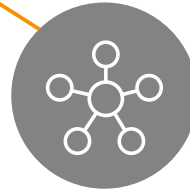
In this session

Introduction
Me, myself and bimforce



Ifc language
While we're fresh: Notepad++!!

Spatial structure
Backbone of Ifc



Modulair design
How you can add, or lose, just about anything

In this session

Locations

Global vs Local placement,
Stacked coordinate systems

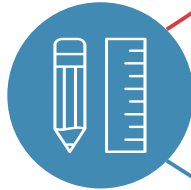


Properties & Attributes



Annotations

Dimensions,
Symbols,
Sections & elevations



Materials

Color definitions,
Material definitions,
Hatches & Textures



In this session

Project library
IFC component libraries



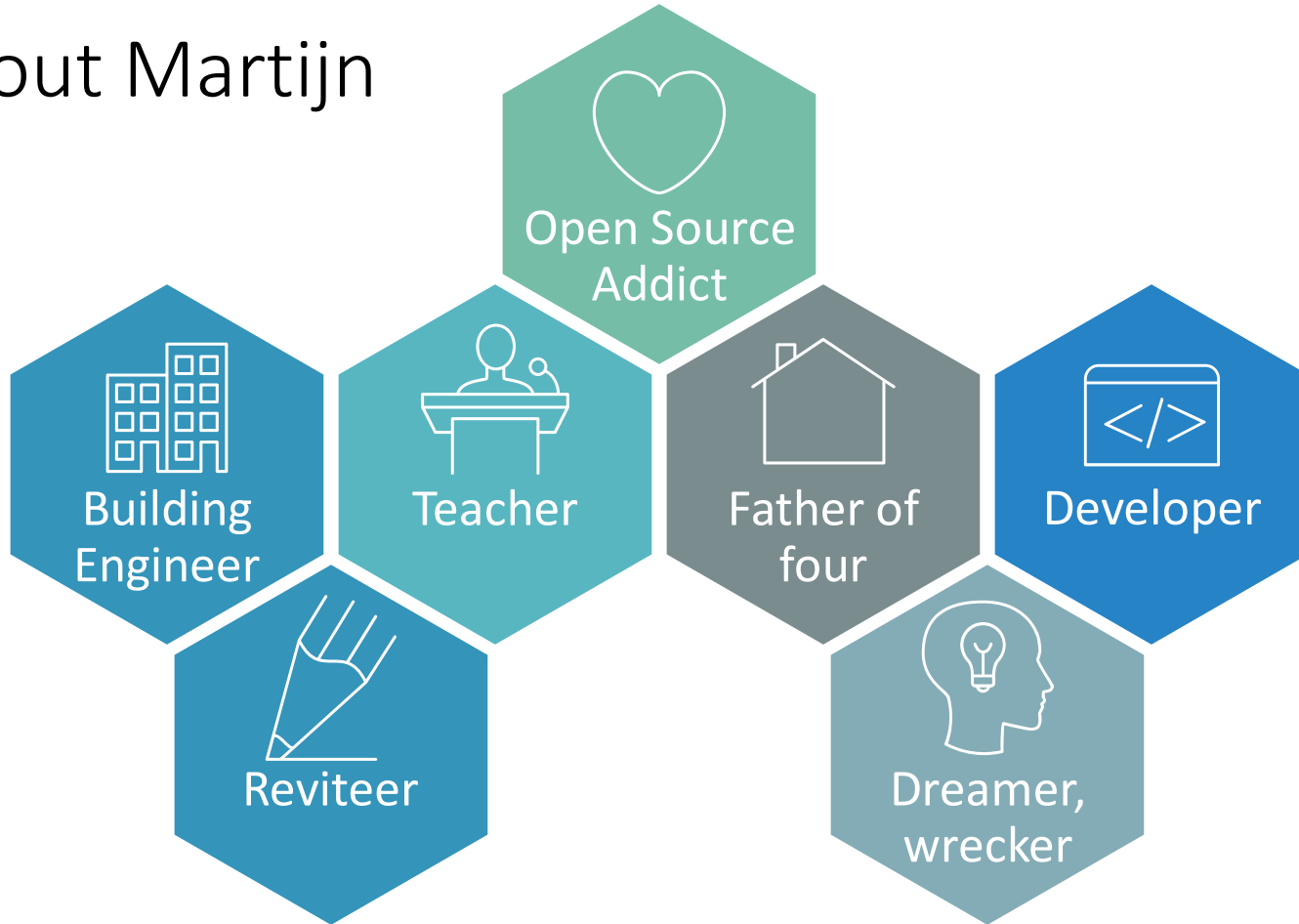
Geometry
How do you define
geometry?

Parametrics
Making stuff flex

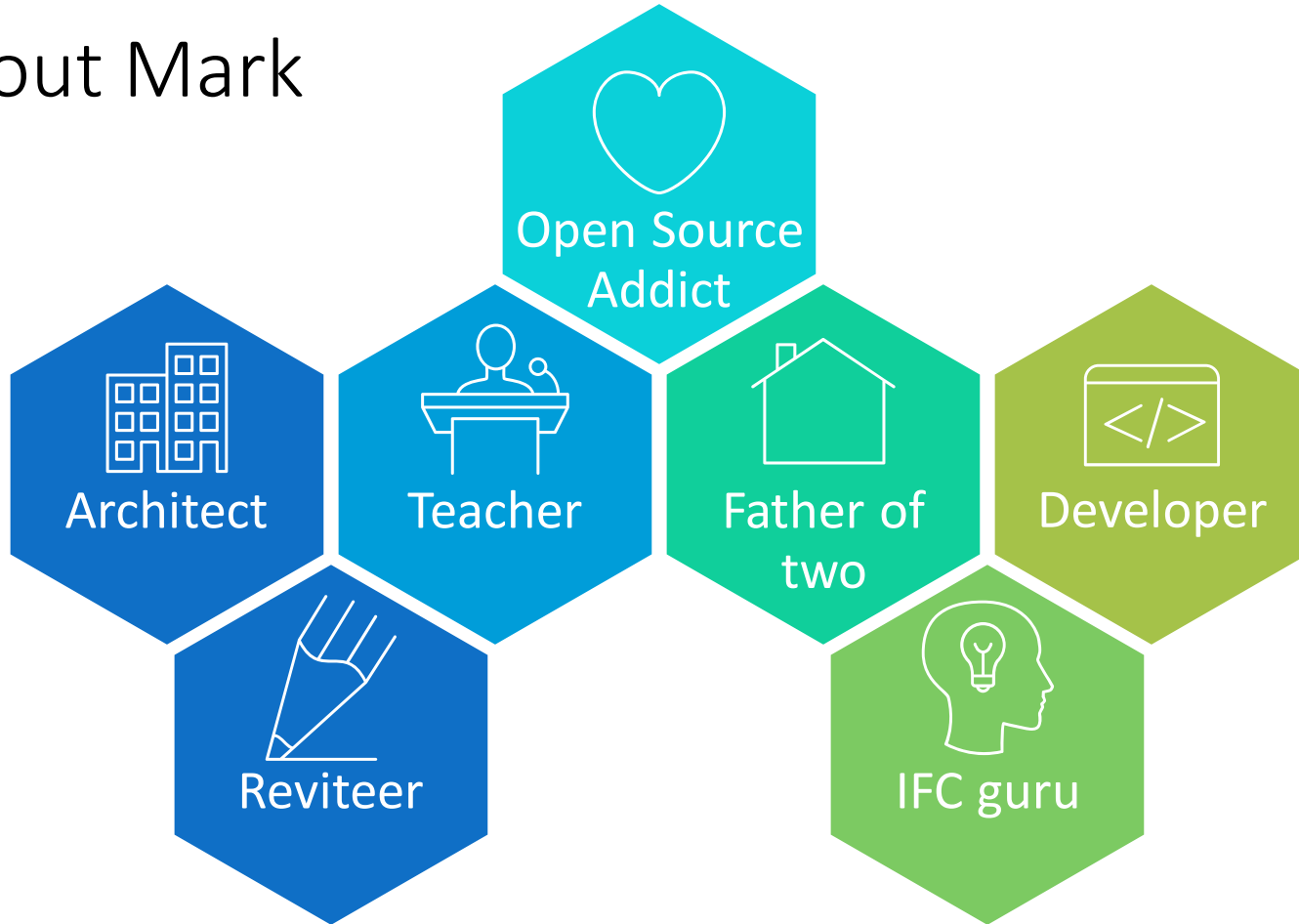


Questions?

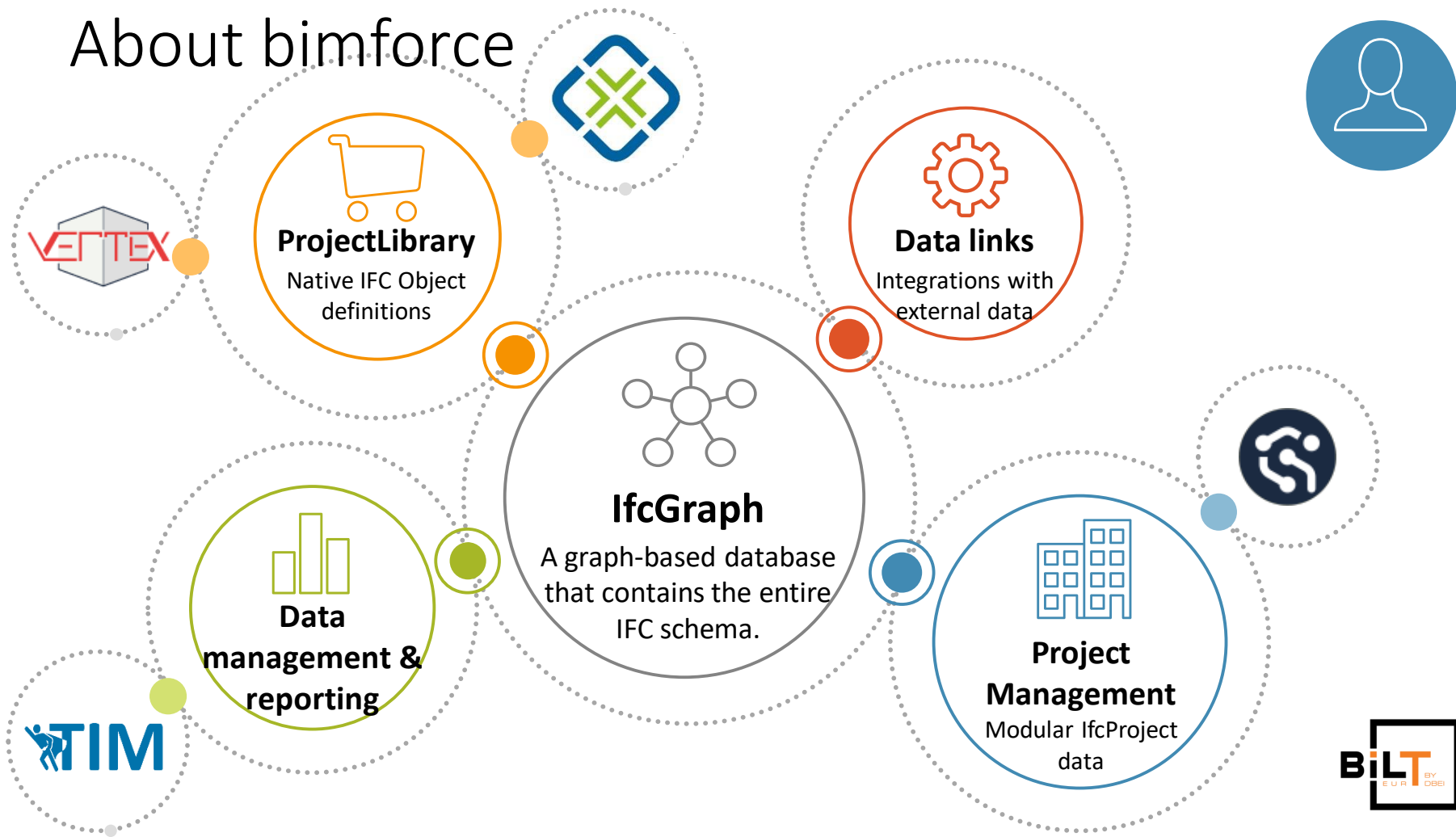
About Martijn



About Mark



About bimforce



About bimforce



GRiDS

- Common Information Environment
- Store all building data in a native IFC Graph database
- Connect with external datasources
- Bidirectional communication with external processes
- Visualisation of data in IFC viewers
- Add information to existing IFCs, extract partials, etc



Vertex

- Native IFC Project Library elements
- Support for full IFC or hybrid Revit solutions
- Support for Families as external geometry placeholder
- Material and Color libraries
- Online Content Management



Housing Configurator

- Fully customizable native IFC housing generator
- 21 housing types with a couple of thousand buyer options and unlimited material options.
- Potential buyers can customize their house with predefined options.
- Automatically creating drawings and contractual documents based on buyer preferences



TIM

- Technical Information Management at your fingertips.
- Define Classifications and add datasets to specific Classifications
- Use Graph Algorithms to automatically scan, review and update IFC models with predescribed information sets.

Ifc language

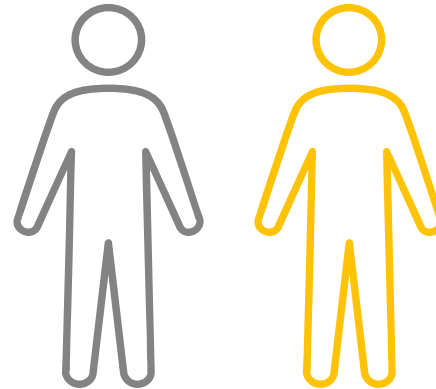
“Ifc is not a file type,
it’s a language”

Common phrase.
What does it mean?

Language

the principal method of
human communication,
consisting of...

...words used in a
structured and
conventional
way and conveyed by
speech, writing, or
gesture.



Ifc language



Technically this means:

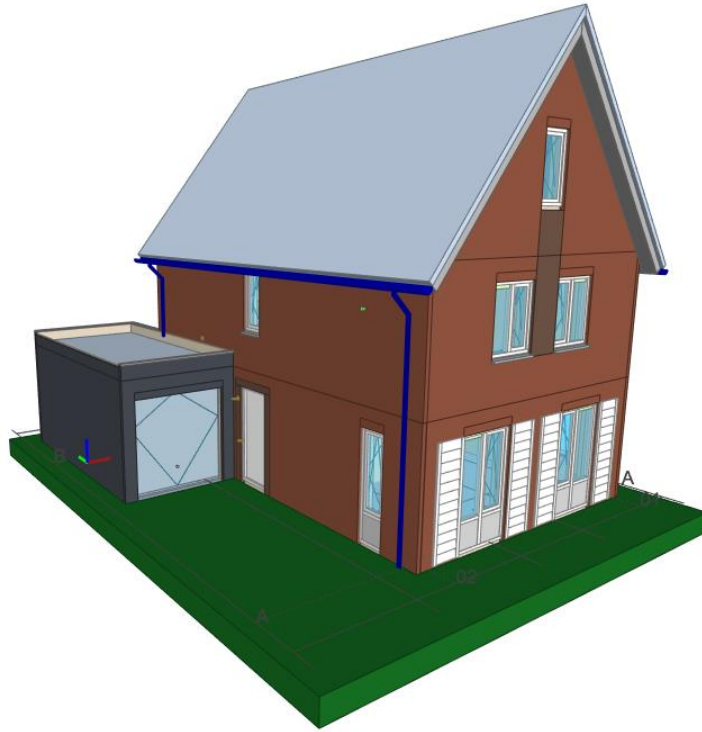


Vocabulary



Grammar

This means it can be written down in multiple ways, and be translated...



Ifc language



No language is perfect. Some of the flaws in the Ifc language are:

- Classifications and Annotations are Instance driven.
- Geometry concepts miss some basic stuff like Arrays, Mirroring

There's no proper way to define several key parts:

- Dimensions
- Annotations types like text or symbols
- Modelviews like a floorplan, elevation, section



IFC EXPRESS = .ifc file

Worst file format ever.

- No regulated file structure.
- Key/value pairs do not have to be in each others' vicinity.
(and can't be)
- Declares everything explicitly

```

3379- IFCPROPERTYSINGLEVALUE('Base Offset',$,IFCLENGTHMEASURE(0.)),);
3380- IFCPROPERTYSINGLEVALUE('Level',$,IFCTEXT('Level 1'),););
3381- IFCPROPERTYSINGLEVALUE('Limit Offset',$,IFCLENGTHMEASURE(5967.37552679513),););
3382- IFCPROPERTYSINGLEVALUE('Upper Limit',$,IFCLABEL('Level: Level 1'),););
3383- IFCPROPERTYSINGLEVALUE('Phase',$,IFCLABEL('New Construction'),););
3384- IFCPROPERTYSINGLEVALUE('Area',$,IFCAREAMEASURE(309.487500000001),););
3385- IFCPROPERTYSINGLEVALUE('Computation Height',$,IFCLENGTHMEASURE(0.)),););
3386- IFCPROPERTYSINGLEVALUE('Perimeter',$,IFCLENGTHMEASURE(70800.00000000001),););
3387- IFCPROPERTYSINGLEVALUE('Unbounded Height',$,IFCLENGTHMEASURE(5967.37552679513),););
3388- IFCPROPERTYSINGLEVALUE('Volume',$,IFCVOLUMEAMEASURE(1733.63532503711),););
3389- IFCPROPERTYSINGLEVALUE('Name',$,IFCTEXT('Gymzaal'),););
3390- IFCPROPERTYSINGLEVALUE('NLR5_C_gebruiks frequentie',$,IFCTEXT(''),););
3391- IFCPROPERTYSINGLEVALUE('NLR5_C_gebruiksfunctie',$,IFCTEXT('sportfunctie'),););
3392- IFCPROPERTYSINGLEVALUE('NLR5_C_gezamenlijk',$,IFCBOOLEAN(F.)),););
3393- IFCPROPERTYSINGLEVALUE('NLR5_C_publiek',$,IFCBOOLEAN(T.)),););
3394- IFCPROPERTYSINGLEVALUE('NLR5_C_vluchtweg',$,IFCBOOLEAN(T.)),););
3395- IFCPROPERTYSINGLEVALUE('NLR5_C_wandafwerking eis',$,IFCTEXT('geschilderd'),););
3396- IFCPROPERTYSINGLEVALUE('Number',$,IFCTEXT('1'),););
3397- IFCPROPERTYSINGLEVALUE('Category',$,IFCLABEL('Rooms'),););
3398- IFCPROPERTYSINGLEVALUE('Phase Id',$,IFCLABEL('New Construction'),););
3399- IFCPROPERTYSET('3_x4$MY4190gX5rn7eC80o',$42,'Constraints',$,{#379,#380,#381,#382});
4401- IFCREDEFINESBYPROPERTIES('3_x4$MY4190gX5rh7eC80o',$42,$,{#331},#399);
4404- IFCPROPERTYSET('3_x4$MY4190gX5r deC80o',$42,'Data',$,{#359});
4406- IFCREDEFINESBYPROPERTIES('3_x4$MY4190gX5rkdC80o',$42,$,{#331},#404);
4409- IFCPROPERTYSET('3_x4$MY4190gX5rVucC80o',$42,'Dimensions',$,{#384,#385,#386,#387,#388});
4411- IFCREDEFINESBYPROPERTIES('3_x4$MY4190gX5rSrgVec80o',$42,$,{#331},#409);
4414- IFCPROPERTYSET('3_x4$MY4190gX5rXlcC80o',$42,'Energy Analysis',$,{#360,#361,#362,#363,#364,#365,#366,#367,#368,#369,#370,#371,
3372,#373,#374,#375,#376,#377,#378});
4416- IFCREDEFINESBYPROPERTIES('3_x4$MY4190gX5rHlcC80o',$42,$,{#331},#414);
4419- IFCPROPERTYSET('3_x4$MY4190gX5rJMcC80o',$42,'Identity Data',$,{#389,#390,#391,#392,#393,#394,#395,#396});
4421- IFCREDEFINESBYPROPERTIES('3_x4$MY4190gX5rSgJcC80o',$42,$,{#331},#419);
4424- IFCPROPERTYSET('1o10EzRw7j0iJm7J6j2w0',$42,'Other',$,{#397,#398});
4426- IFCREDEFINESBYPROPERTIES('2yHk10iCvD4A9uKU0LzrT',$42,$,{#331},#424);
4429- IFCPROPERTYSET('3_x4$MY4190gX5rVucC80o',$42,'Phasing',$,{#383});

```

What are the other options?

Ifc language in IfcJSON



- Makes IFC more “readable” by solving part of structure issue.
- Still not able to express networked IFC data structure
- We use it a lot, translator provided by GeometryGym

```
1 {
2   "HEADER": {
20    "DATA": {
21      {
22        "Name": "Modufair Woningconfigurator - Woningtype-D4",
23        "type": "IfcProject",
24        "LongName": "Jansen de Jong - Woningconfigurator Woningtype D4",
25        "GlobalId": "23ola0WXv3BgaQtLOxCiBU",
26        "Phase": "Kopersopties",
27        "IsDefinedBy": {
110      "RepresentationContexts": {
245      "UnitsInContext": {
282      "IsDecomposedBy": {
283        {
284          "type": "IfcRelAggregates",
285          "RelatedObjects": [
286            {
287              "type": "IfcSite",
288              "Name": "Kavel 5",
289              "GlobalId": "1N7ijwmrj3lBOFjYW$CSPl",
290              "CompositionType": "ELEMENT",
291              "IsDefinedBy": {
374      "ObjectPlacement": {
387      "IsDecomposedBy": {
388        {
389          "type": "IfcRelAggregates",
390          "RelatedObjects": [
391            {
392              "ElevationOfRefHeight": "0",
393              "Name": "JDJ - VerteX Project",
394              "ElevationOfTerrain": "0",
395              "LongName": "Jansen de Jong - Woningconfigurator Woningtype D4 - Kavel 5",
396              "Description": "Woning configuratie: Klant nr.0001",
397              "CompositionType": "ELEMENT",
398              "ObjectType": "Hoekwoning Type-D4",
399              "GlobalId": "0AMi2C6x9ly9Gy24ZcKbkv",
400              "type": "IfcBuilding",
401              "IsDefinedBy": {
460      "ObjectPlacement": {
476      "IsDecomposedBy": {
```

Ifc language in IfcGraph



IfcGraph:

- Makes IFC more “readable” by solving part of structure issue.
- Allows for full usage of the networked IFC language structure

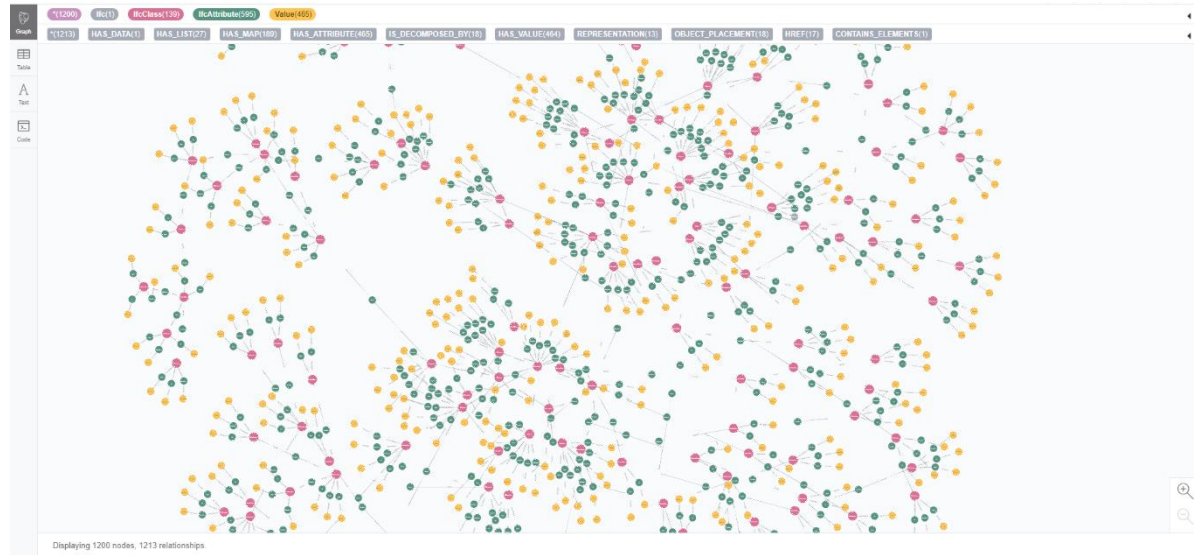
Allow me to demonstrate...

Ifc language in IfcGraph



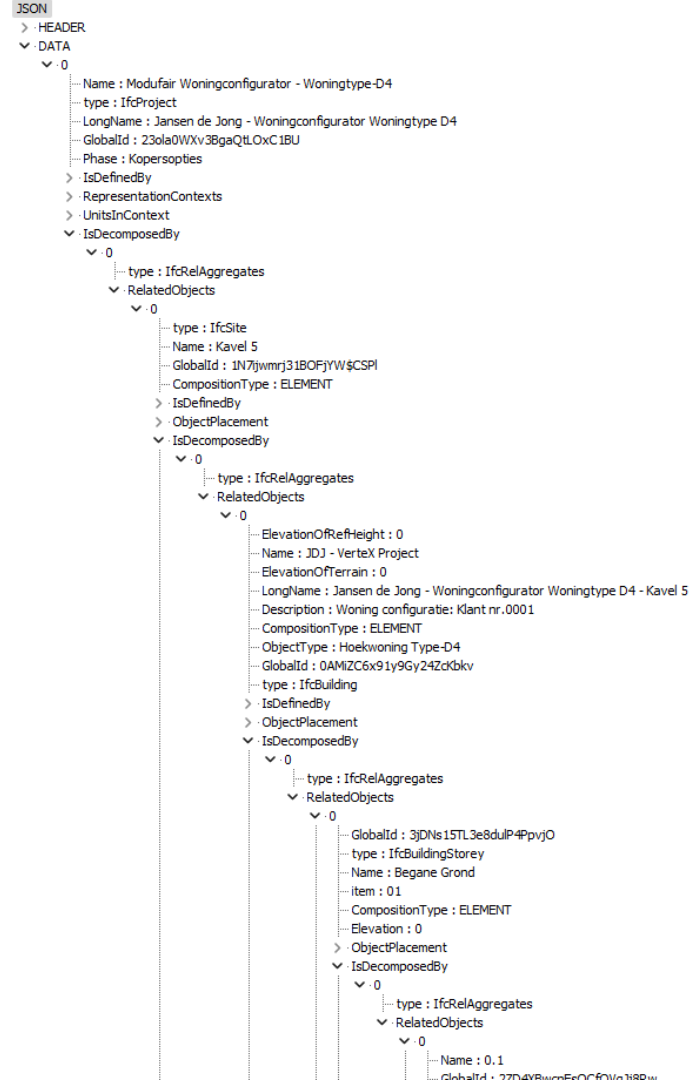
IfcGraph:

This is roughly 12% of a simplified version of the Ifc dataschema in a Graph database for a single house



Spatial structure in IfcJSON

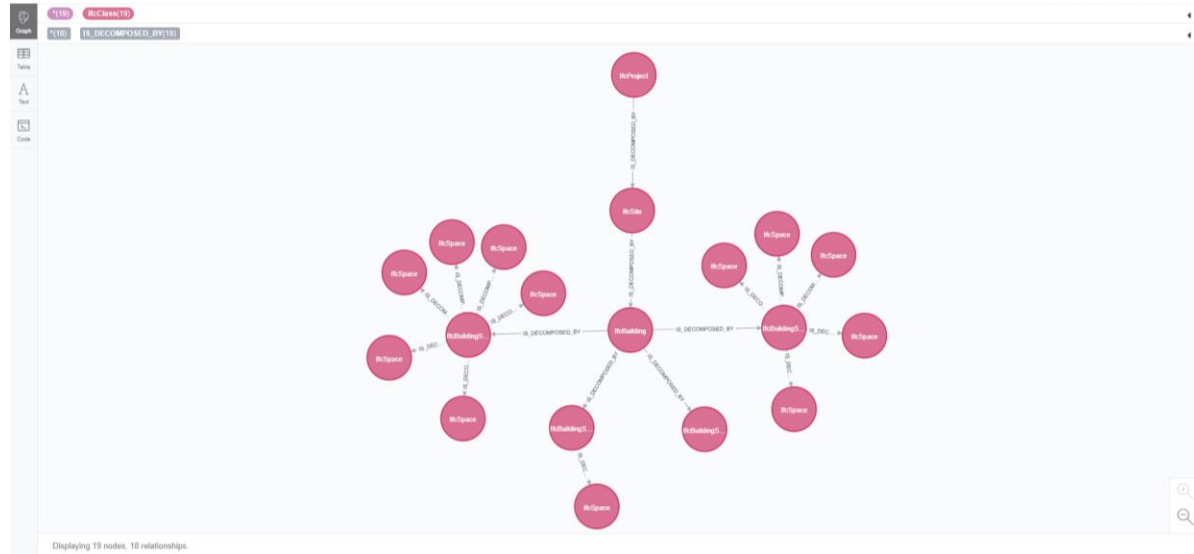
- The Spatial Structure defines the first steps in the Ifc.
- It contains:
 - IfcProject
 - IfcSite
 - IfcBuilding
 - IfcBuildingStorey
 - IfcSpace
- Defined by IsDecomposedBy relationship



Spatial structure in IfcGraph



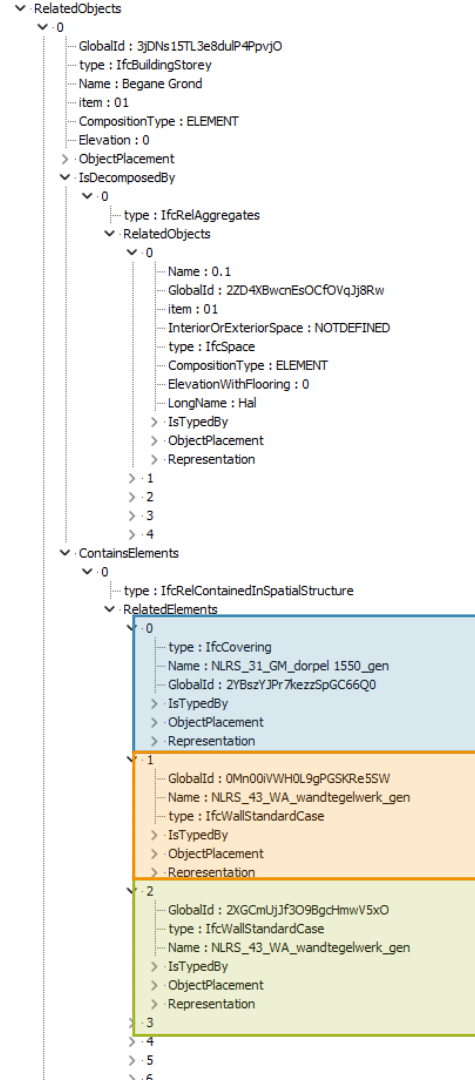
- The Spatial Structure defines the first steps in the Ifc.
- It contains:
 - IfcProject
 - IfcSite
 - IfcBuilding
 - IfcBuildingStorey
 - IfcSpace
- Defined by IsDecomposedBy relationship



Spatial structure in IfcJSON

- Other objects are hosted on the Spatial Structure using an ContainsElements relationship.
- Image shows first added component to Building Storey “Begane grond”:
 - IfcCovering
 - IfcWallStandardCase
 - IfcWallStandardCase

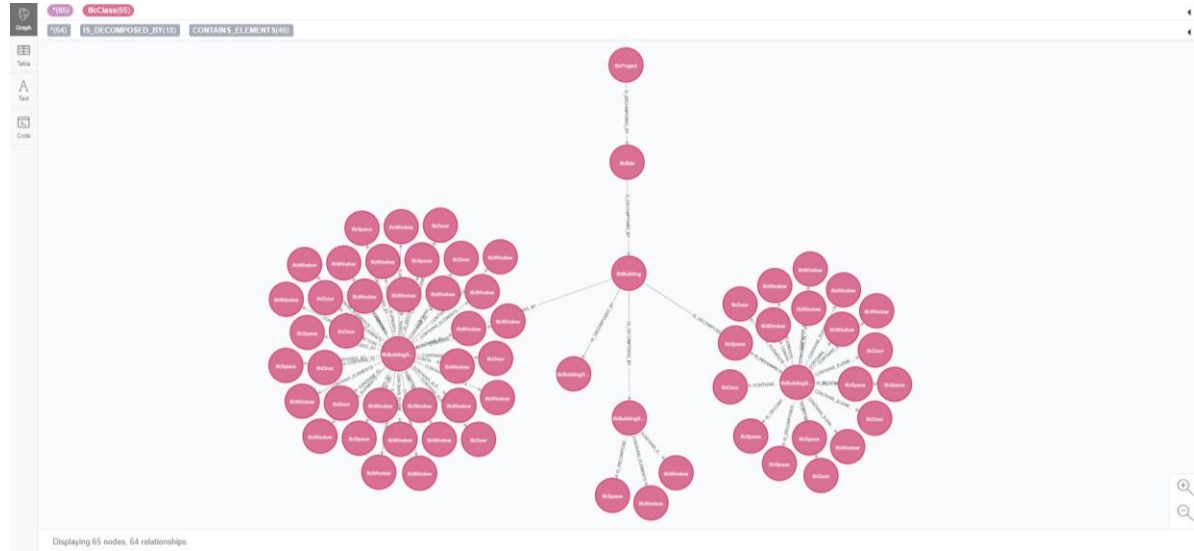
Still rather straightforward



Spatial structure in IfcGraph



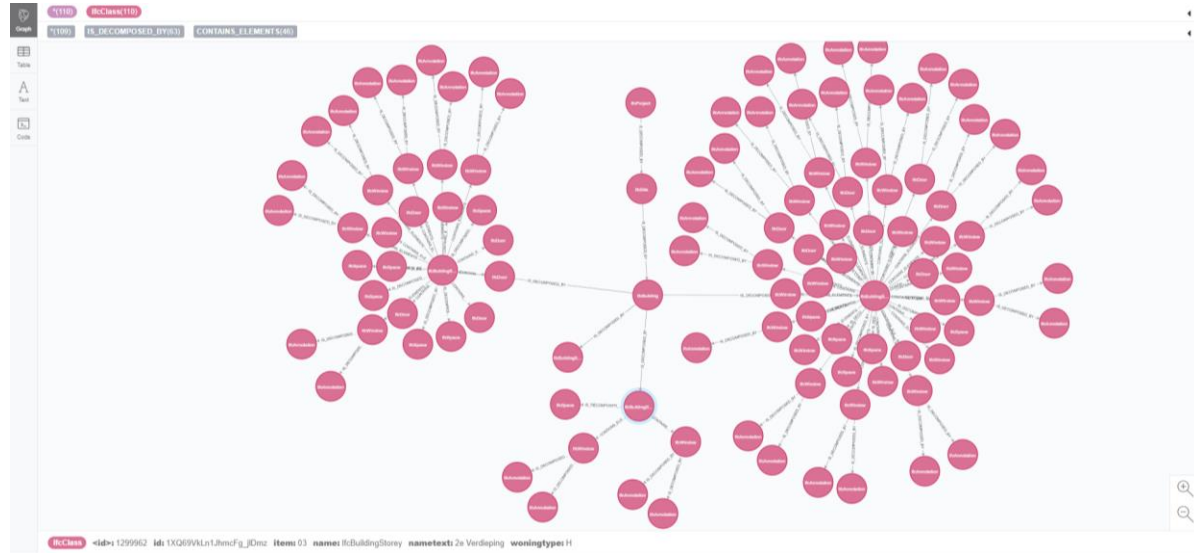
- Other objects are hosted on the Spatial Structure using an IsContainedIn relationship.
- Image shows IfcWindows and IfcDoors, added to Ground floor and First and Second Floor Building Storeys



Spatial structure in IfcGraph



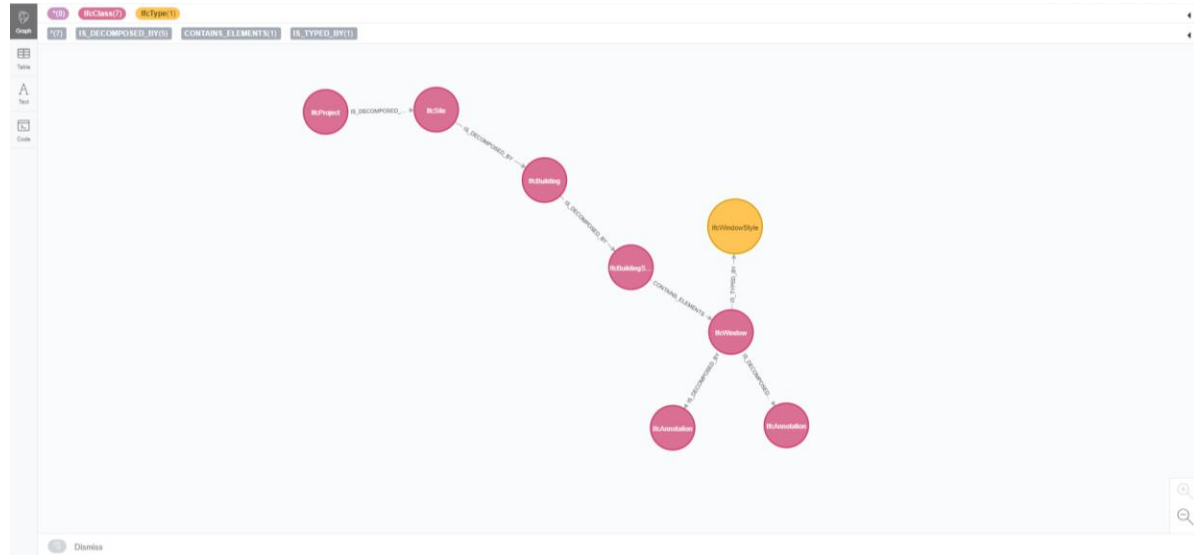
- Elements can be broken down into sub-components.
- In this example you can see IfcAnnotations (e.g. swing directions) as part of the IfcDoors.



Spatial structure



- In more detail the entire branche leading to a specific IfcWindow instance,
- With it's attached swing symbols as IfcAnnotations
- And a reference to it's IfcType



Modular design



Which brings us to an important part:

Modularity within the Ifc schema.

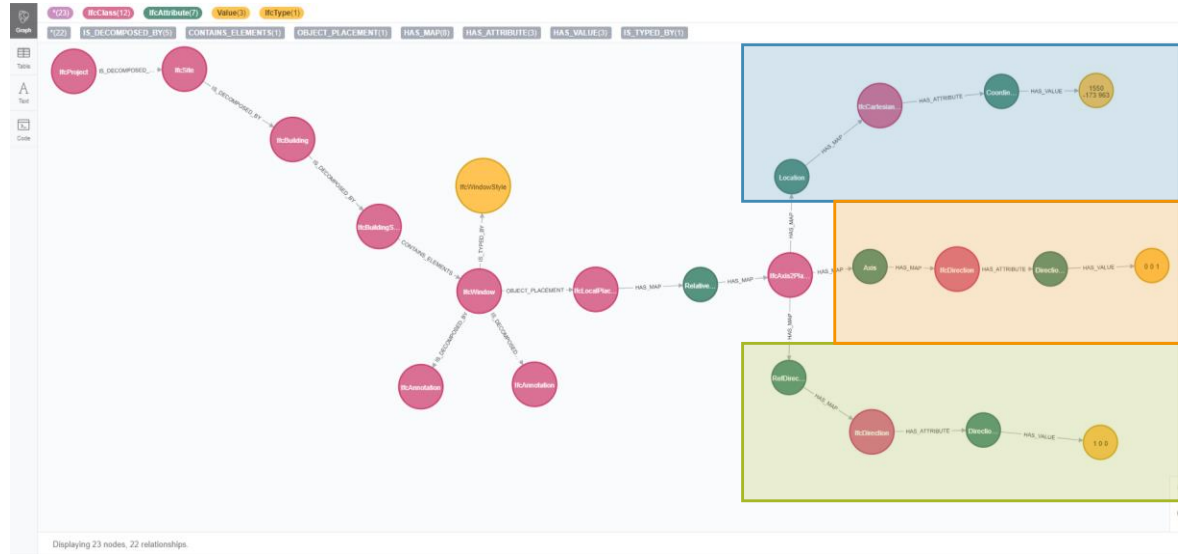
After implementing the IfcEntity schema you can effectively edit and update any Ifc file attached to your database.

In any granularity you want.

Let's discuss some “Modules”

Locations

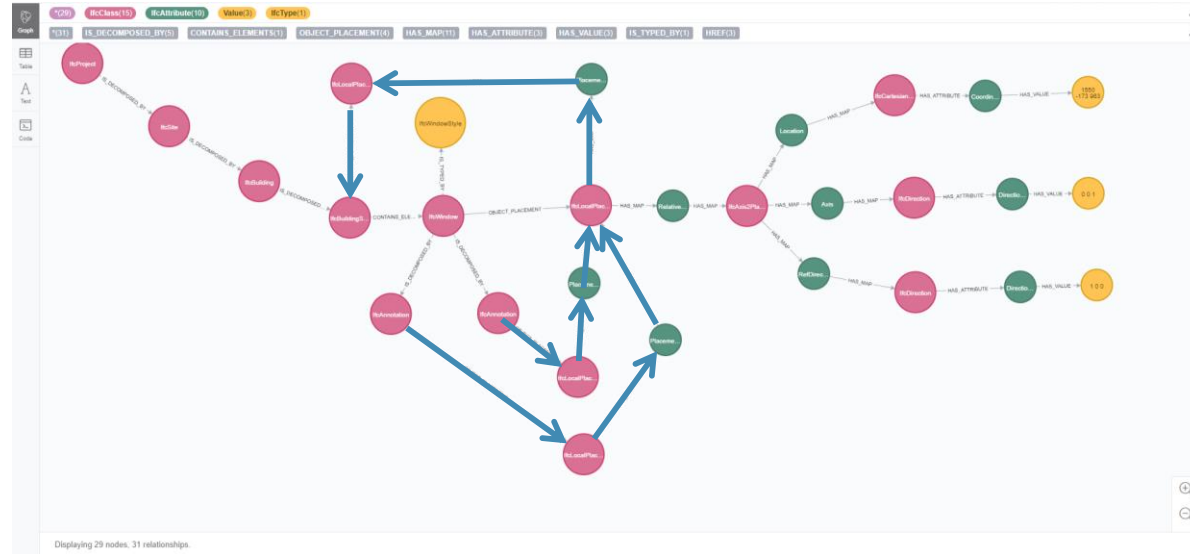
- Each instance has a Local Placement.
- This consists of three things:
 - Coordinates
 - Axis
 - Direction



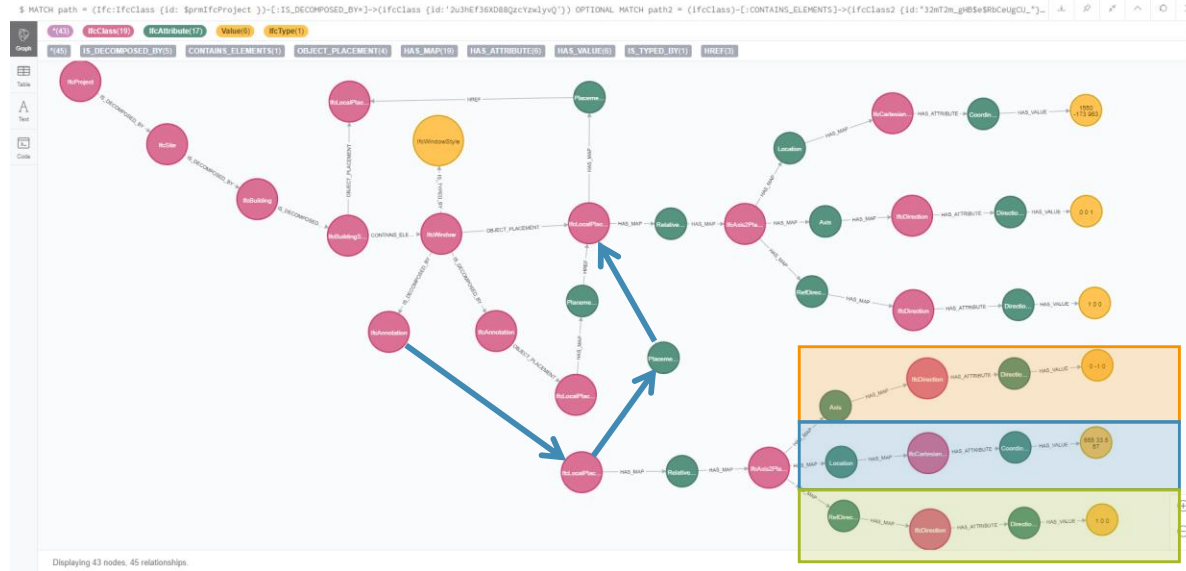
Locations



- But can also be hosted relatively to another object.
- In this example the IfcWindow also has a reference to:
 - The two Annotations
 - The Building Storey



Axis, Direction and Coordinates



Locations



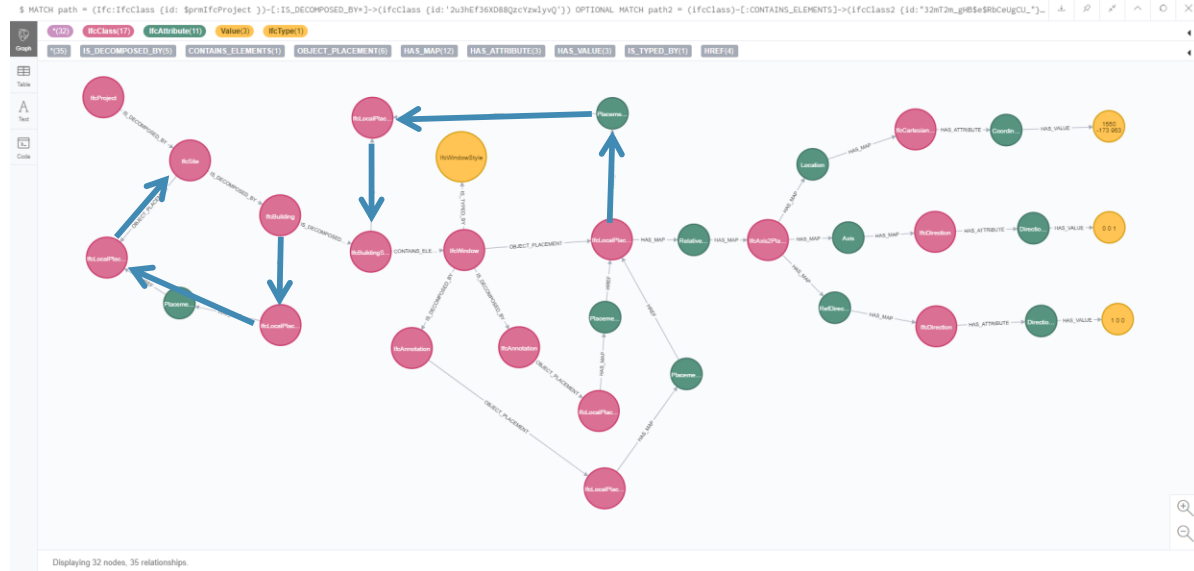
Relative locations can be stacked on top of each other:

The Annotation can be placed relative to the Window,

Which is placed relative to the BuildingStorey,

Which is placed relative to the Building,

Which is placed relative to the Site



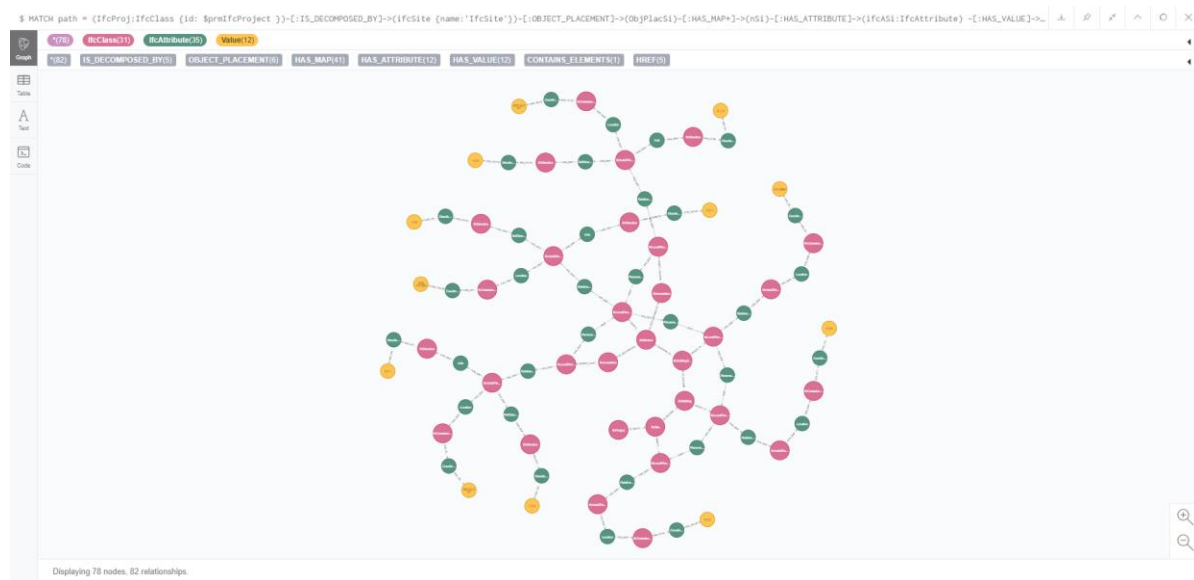
Locations



The other option is to ditch relative placements and only work with absolute (global) placement.

That would make it a lot easier (and is what most exporters do)

But it would also be less fun.

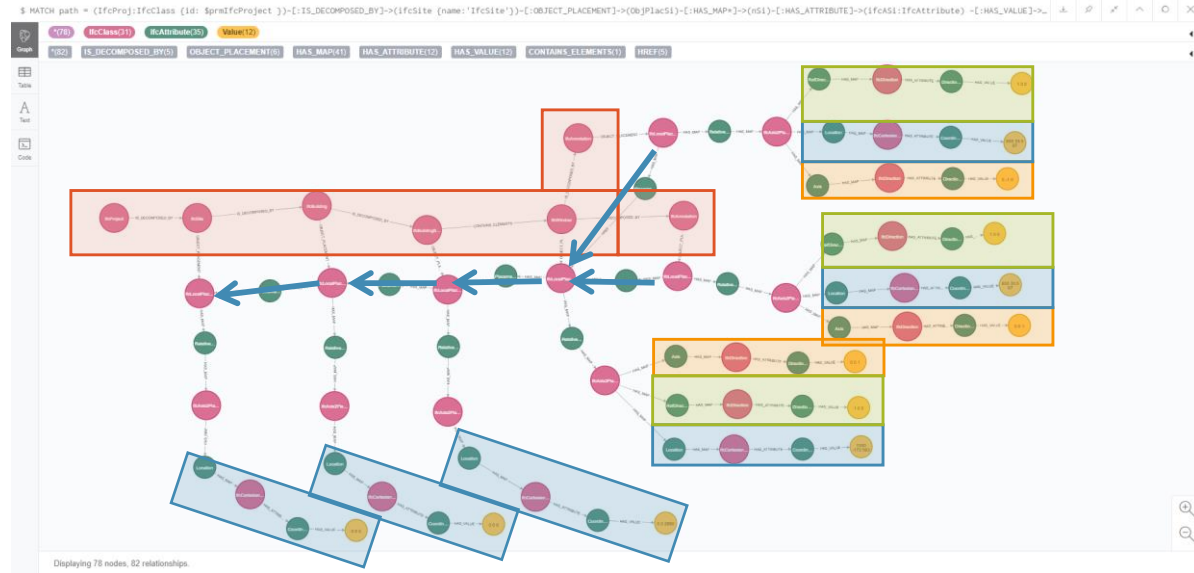


Locations



A single object can be hosted relative to a chain of other components.

In this example two Annotations, with their own location, are placed relative to a stack of 4 coordinates, and 1 direction/rotation



- | Industry Foundation Classes Release 4 (IFC4) | | | |
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| F. Change logs | | | |
| F.1 IFC4 | | | |
| F.1.1 Change log | | | |
| | id=PresentationResource | X X | DELETED |
| | IfcProductOfCombustionProperties | X X | DELETED |
| | IfcRetraction | X X | DELETED |
| | IfcThermalMaterialProperties | X X | DELETED |
| | IfcTopicProperties | X X | DELETED |
| | IfcPRESENTATIONDIMENSIONINGRESOURCE | | DELETED |
| | IfcDimensionExtentUsage | | DELETED |
| | IfcDraftingCalloutElement | | DELETED |
| | IfcAngularDimension | X X | DELETED |
| | IfcDiameterDimension | X X | DELETED |
| | IfcDimensionCalloutRelationship | X X | DELETED |
| | IfcDimensionCurve | X X | DELETED |
| | IfcDimensionCurveDirectedCallout | X X | DELETED |
| | IfcDimensionCurveTerminator | X X | DELETED |
| | IfcDimensionPair | X X | DELETED |
| | IfcDraftingCallout | X X | DELETED |
| | IfcDraftingCalloutRelationship | X X | DELETED |
| | IfcLinearDimension | X X | DELETED |
| | IfcPreDefinedDimensionSymbol | X X | DELETED |
| | IfcPreDefinedPointMarkerSymbol | X X | DELETED |
| | IfcPreDefinedTerminatorSymbol | X X | DELETED |
| | IfcProjectionCurve | X X | DELETED |
| | IfcRadiusDimension | X X | DELETED |
| | IfcStructureDimensionCallout | X X | DELETED |
| | IfcTerminatorSymbol | X X | DELETED |
| | IfcPRESENTATIONRESOURCE | | DELETED |
| | IfcFontStyle | MOVED | Schema changed from IfcPresentationResource to IfcPresentationAppearanceResource. |
| | IfcFontVariant | MOVED | Schema changed from IfcPresentationResource to IfcPresentationAppearanceResource. |
| | IfcFontWeight | MOVED | Schema changed from IfcPresentationResource to IfcPresentationAppearanceResource. |
| | IfcFontStyleText | MOVED | Schema changed from IfcPresentationResource to IfcPresentationAppearanceResource. |
| | IfcTextFontName | MOVED | Schema changed from IfcPresentationResource to IfcPresentationAppearanceResource. |
| | IfcColour | MOVED | Schema changed from IfcPresentationResource to IfcPresentationAppearanceResource. |
| | IfcTextFontSelect | MOVED | Schema changed from IfcPresentationResource to IfcPresentationAppearanceResource. |
| | IfcColourRgb | MOVED | Schema changed from IfcPresentationResource to IfcPresentationAppearanceResource. |
| | IfcColourSpecification | MOVED | Schema changed from IfcPresentationResource to IfcPresentationAppearanceResource. |

Attributes



There are a two Attributes you absolutely need:

- Name
- GlobalId

Other Attributes define relationships. For example:

- Object Type
- PredefinedType

These Attributes explicitly state the endpoint of a defined type of relationship.

IFC4_ADD2_TC1 - 4.0.2.1 [Official]

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6. Shared element data schemas

Attribute inheritance

#	Attribute	Type	Cardinality	Description	
1	GlobalId	IfcGlobalId	1	Assignment of a globally unique identifier within the entire software world.	[Standard list]
2	OwnerHistory	IfcOwnerHistory	1	Assignment of the information about the current ownership of that object, including owning actor, application, local identification and information captured about the recent changes of the object. <i>IFC4 COMMENT: This attribute has been deprecated in IFC4. It is not recommended to use it.</i>	
3	Name	IfcText	1	Optional name for use by the participating software systems or users. For some subtypes of IfcText the inclusion of the Name attribute may be required. This would be enforced by a where rule.	
4	Description	IfcText	1	Optional description, provided for exchanging informative comments.	
IfcObjectDefinition					
	HasAssignments	IfcAssignments @PartOfObjects	0..1	Reference to the relationship objects, that assign (by an association relationship) other subtypes of IfcObject to this object instance. Examples are the association to products, processes, controls, resources or groups.	
	Has	IfcObject @PartOfObjects	0..1	Reference to the decomposition relationship being a nesting. It determines that this object definition is a part within an ordered whole/part decomposition relationship. An object occurrence or type can only be part of a single decomposition (to allow hierarchical structures only). <i>IFC4 COMMENT: This attribute has been deprecated in IFC4. It is not recommended to use it.</i>	
	IsNestedBy	IfcObject @PartOfObjects	0..1	Reference to the decomposition relationship being a nesting. It determines that this object definition is the whole within an ordered whole/part decomposition relationship. An object or object type can be nested by several other objects (occurrences or types). <i>IFC4 COMMENT: This attribute has been deprecated in IFC4. It is not recommended to use it.</i>	
	HasContext	IfcContext @PartOfObjects	0..1	Reference to the context providing context information such as project unit or representation context. It should only be asserted for the uppermost non-spatial object.	
	IsCompositedBy	IfcObject @PartOfObjects	0..1	Reference to the decomposition relationship being an aggregation. It determines that this object definition is a whole within an unordered whole/part decomposition relationship. An object definition can be aggregated by several other objects (occurrences or parts).	
	Decomposes	IfcObject @PartOfObjects	0..1	Reference to the decomposition relationship being an aggregation. It determines that this object definition is a part within an unordered whole/part decomposition relationship. An object definition can only be part of a single decomposition (to allow hierarchical structures only). <i>IFC4 COMMENT: This attribute has been deprecated in IFC4. It is not recommended to use it.</i>	
	HasAssociations	IfcObject @PartOfObjects	0..1	Reference to the relationship objects, that associates external references or other resource definitions to this object. Examples are the association to library, documentation or classification.	
IfcObject					
5	ObjectType	IfcText	1	The type denotes a particular type that instantiates the object further. The use has to be established at the level of instantiable subtypes. In particular it holds the user defined type, if the enumeration of the attribute PredefinedType is set to USERDEFINED.	
	IsCreatedBy	IfcObject @PartOfObjects	0..1	Link to the relationship object pointing to the creating object that provides the object definitions for the object occurrence. The creating object has to be part of an object type decomposition. The associated IfcObject, or its subtypes, contains the specific information (as part of a type, or style, definition), that is common to all related instances of the creating IfcObject, or its subtypes. <i>IFC4 COMMENT: This attribute has been deprecated in IFC4. It is not recommended to use it.</i>	
	Declares	IfcObject @PartOfObjects	0..1	Link to the relationship object pointing to the reflected object(s) that specifies the object definitions. The reflected object has to be part of an object occurrence decomposition. The associated IfcObject, or its subtypes, provides the specific information (as part of a type, or style, definition), that is common to all related instances of the declaring IfcObject, or its subtypes.	
	IsTypically	IfcObject @PartOfObjects	0..1	Set of relationships to the object type that provides the type definitions for the object occurrence. The associated IfcObject, or its subtypes, contains the specific information (as part of a type, or style, definition), that is common to all instances of IfcObject, or its subtypes, referring to the same type. <i>IFC4 COMMENT: This attribute has been deprecated in IFC4. It is not recommended to use it.</i>	
	IsCreatedBy	IfcObject @PartOfObjects	0..1	Set of relationships to property set definitions attached to this object. These statically or dynamically defined properties contain algorithmic information content that further defines the object. <i>IFC4 COMMENT: This attribute has been deprecated in IFC4. It is not recommended to use it.</i>	
IfcProduct					
6	ObjectPlacement	IfcObjectPlacement	1	Placement of the product in space, the placement can either be absolute (relative to the world coordinate system), relative (relative to the object placement of another product), or constant (e.g. relative to grid axes). It is determined by the various subtypes of IfcObjectPlacement, which includes the area placement information to determine the transformation for the object coordinate system.	
	Representation	IfcProductRepresentation	1	Reference to the representations of the product, being either a representation (IfcProductRepresentation) or as a special case a shape representation (IfcProductRepresentationShape). The product definition shape provides for multiple geometric representations of the shape property of the object within the same object coordinate system, defined by the object parameter.	
	ReferencesBy	IfcProductRepresentation	0..1	Reference to the IfcProductRepresentation relationship, by which other products, processes, controls, resources or actors (as subtypes of IfcObjectDefinition) can be related to this product.	
IfcElement					
7	Tag	IfcText	1	The tag (or label) identifier at the particular instance of a product, e.g. the serial number, or the position number. It is the identifier at the occurrence level.	

Property Sets in IfcJSON

PropertySets can be declared on the IfcType and Instance.

They basically work the same as Revit Type and Instance parameters.

```
... type : IfcSite
... Name : 5
... GlobalId : 1N7jwmrj31BOFYjYw$CSPi
... CompositionType : ELEMENT
... IsDefinedBy
  ... 0
    ... RelatingPropertyDefinition
      ... GlobalId : 2U_dVY7SL0YwZnMl3B16A
      ... type : IfcPropertySet
      ... Name : Pset_IDJ_Eigenschappen
      ... HasProperties
        ... 0
          ... Name : Woningtype
          ... type : IfcPropertySingleValue
          ... id : 3QQQWeAs4f6QRRXgN5PipXy
          ... NominalValue
            ... IfcLabel : Woningtype D4
        ... 1
          ... id : 3gUsS78K51hwnH8WQPr$u
          ... type : IfcPropertySingleValue
          ... Name : Kadastraal adres
          ... NominalValue
            ... IfcLabel : Gemeente, Sectie, Nummer
        ... 2
          ... Name : Huisnummer
          ... type : IfcPropertySingleValue
          ... id : 1GdK9Aav5LvYkohvtJEuN
          ... NominalValue
            ... IfcLabel : Huisnummer
        ... 3
          ... Name : Postcode
          ... id : 1FMRClApD8vwx8m39NJU8
          ... type : IfcPropertySingleValue
          ... NominalValue
            ... IfcLabel : Postcode
        ... 4
          ... type : IfcPropertySingleValue
          ... id : 2KNrzdCT12P$UPwN3$XJN
          ... Name : Huisnummer Toevoeging
          ... NominalValue
            ... IfcLabel : Toevoeging
        ... 5
          ... id : 320wVxB7P9eurNo$gf0Pwm
          ... type : IfcPropertySingleValue
          ... Name : Woonplaats
          ... NominalValue
            ... IfcLabel : Plaats
        ... 6
          ... Name : Kavelnummer
          ... id : 2VPYHNFxn3bfbSXhSP2GQV
          ... type : IfcPropertySingleValue
          ... NominalValue
            ... IfcInteger : 1
        ... 7
          ... type : IfcPropertySingleValue
          ... Name : Straat
          ... id : 0HdYvxFdL4KpQ4P185ph5
```

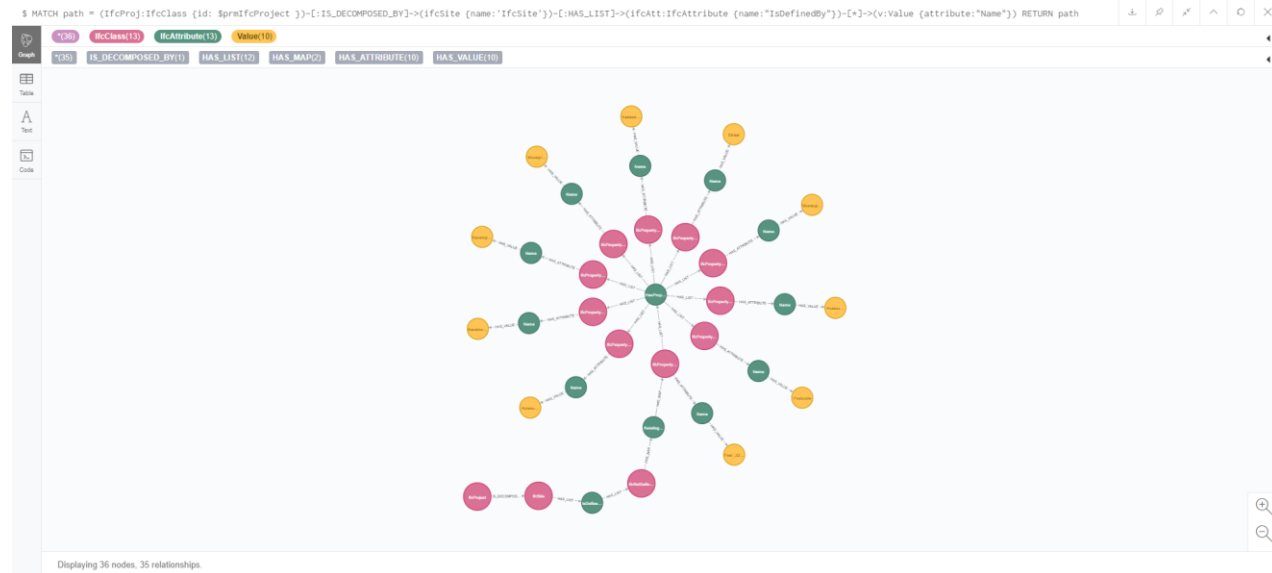
Property Sets in IfcGraph



PropertySets can be retrieved with a single set.

These are explicitly defined in “full IFC mode”.

It allows us to directly manipulate any IFC propertyset data



Materials



- Materials in IFC have a bad reputation.
- This is completely unjustified.
- IfcMaterials can hold any information that (for instance) Revit does:
 - Colors
 - Render Textures
 - Bump maps
 - Material Properties
 - And so on...

To demonstrate this, some of the work we have been doing on Materials

Project Library



- Element Instances are based on an IfcType
- These can be defined in a project library.
This works similarly to Revit:
 - Properties can be type or instance
 - Geometry is Type based
 - Layered structures are an exception. These types, much like Revit, do not have type-based geometry.
 - The same applies to IfcAnnotations
 - Elements can be (shared!) nested components.



Geometry

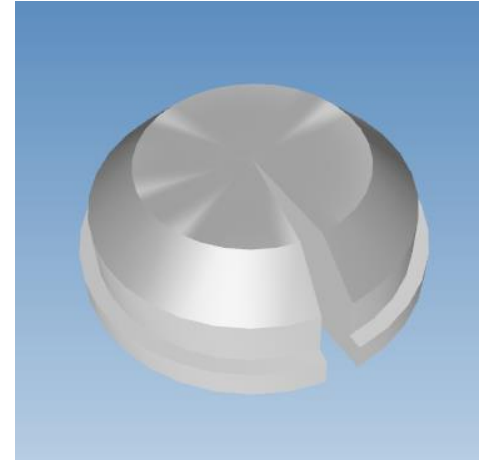
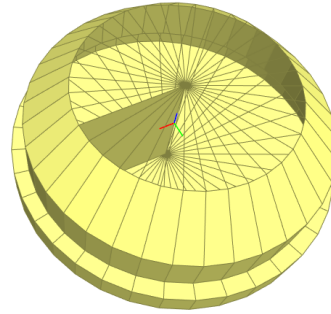


- IFC allows you to simultaneously use multiple ways of defining objects.
- So Vertex uses:
 - [Full IFC definitions for all Materials](#)
 - [Full IFC definitions for all Layered Structures](#)
 - It CAN use IFC geometry for every other component too.
 - While Revit Families are also supported as “geometry placeholders”

Geometry

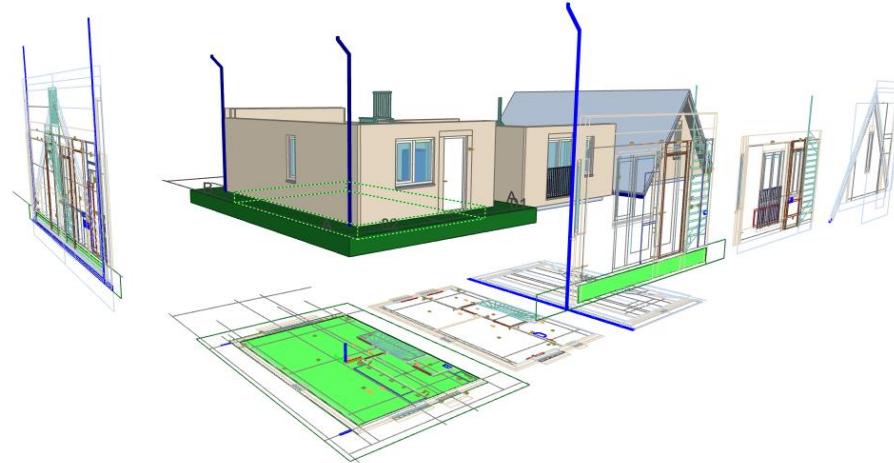
- IFC4 holds a lot of concepts in defining geometry.
- However, there are issues. Mostly with implementation.

This is a revolve in several viewers. Notice how they all “translate” it differently.



Geometry

- Some viewers simply create meshes from everything.
- Others do a better job at preserving “intelligent” ways of describing geometry



Parametrics



- Two major issues with Parametrics in IFC:

1. It doesn't support "Formulas".
We solved this outside of the schema.

2. The schema is incomplete. Not all objects have GlobalIds to reference. Mainly: IfcAttributes and relationships.

```
{  
  "CompositionType": "ELEMENT",  
  "GlobalId": "2w6z7hhz19Eg8vtX_FlOLn",  
  "type": "IfcBuildingStorey",  
  "Name": "1e Verdieping",  
  "item": "02",  
  "Elevation": "2880",  
  "ObjectPlacement": {  
    "id": "2oWaAgkBT0Qhpsx2waj9lP",  
    "type": "IfcLocalPlacement",  
    "RelativePlacement": {  
      "id": "24_mrtzQTC7g2KrPBM9Lvs",  
      "type": "IfcAxis2Placement3D",  
      "Location": {  
        "type": "IfcCartesianPoint",  
        "id": "1B4XbAU9rAOAA7nu$zQOV9",  
        "Coordinates": "0 0 3500"  
      }  
    }  
  }  
}
```

IfcGraph does support this.

Wrap-up

- IFC is a datastructure, treat it as such
- It is not especially “friendly” for file-based usage
- However, using it as a Graph works surprisingly well
- There literally is nothing you cannot model in IFC that you can model in Revit.

Except an array.

One on One Meeting

I will be at the **Speaker Lounge** in the Exhibition Hall for further conversations.

Please join me there after the presentation if you have any further questions

REMINDER:

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Fill the Survey in the Event App





IFC for Smartie-pants

Martijn de Riet & Mark Wieringa

Bimforce

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UNIVERSITAT POLITÈCNICA DE VALÈNCIA

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14 – 16 NOV 2022



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IFC Housing Configurator: Driving building automation with IFC



Mark Wieringa & Martijn de Riet

Bimforce



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In this session

Introduction

Me, myself and bimforce



Client question

What were the project specifications



Configuring a project
Demonstration of setting up a project

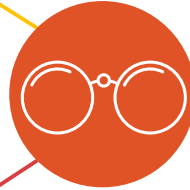


IfcGraph

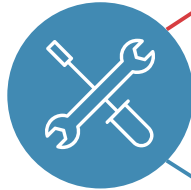
Developing an IfcProject(Library)
Graph

In this session

Trials and Tribulations
Major issues to overcome



**Developing Open Source
IFC Viewer**
With IfcOpenShell & Three.js

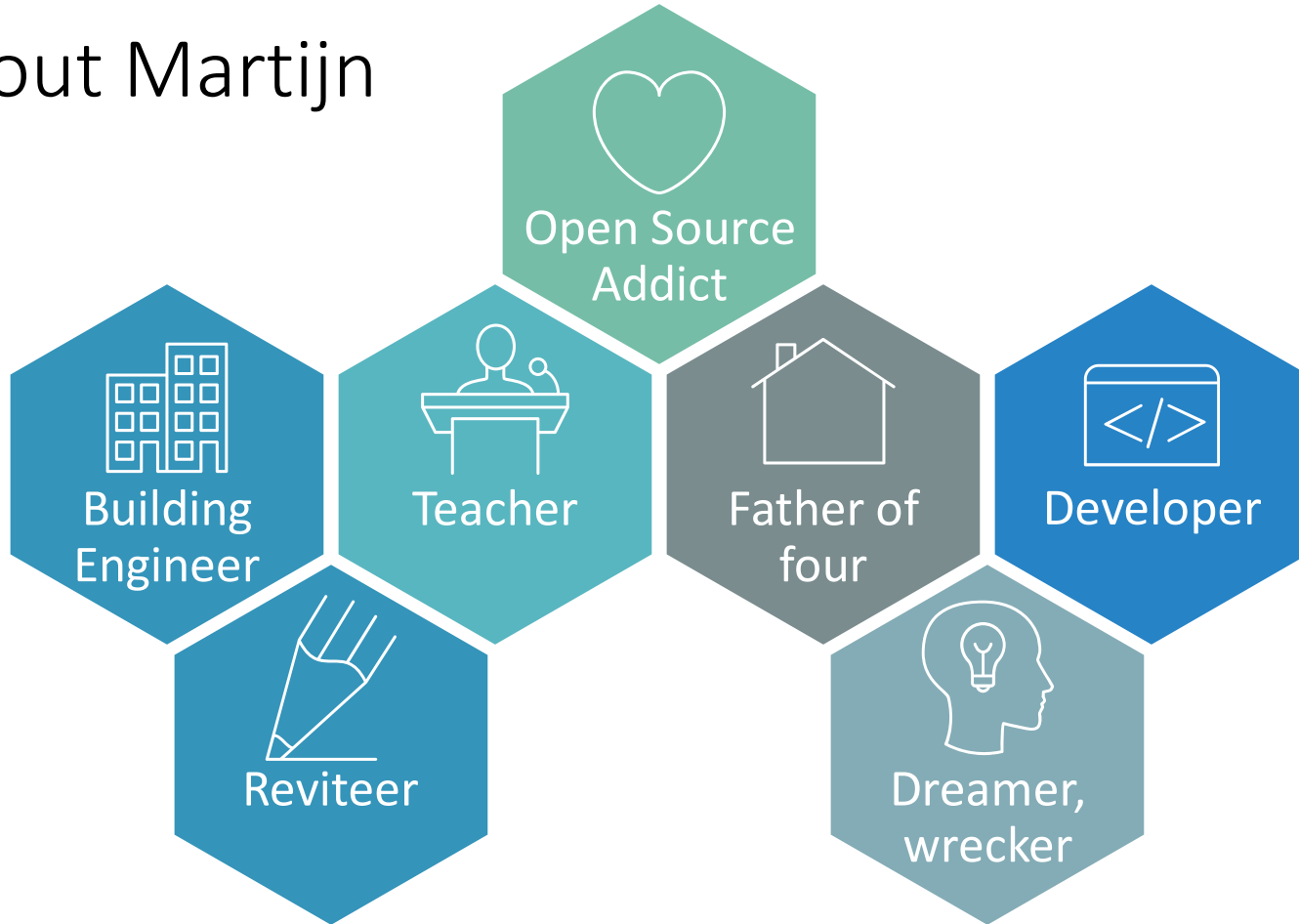


Building blocks
Implementing IfcProjectLibrary

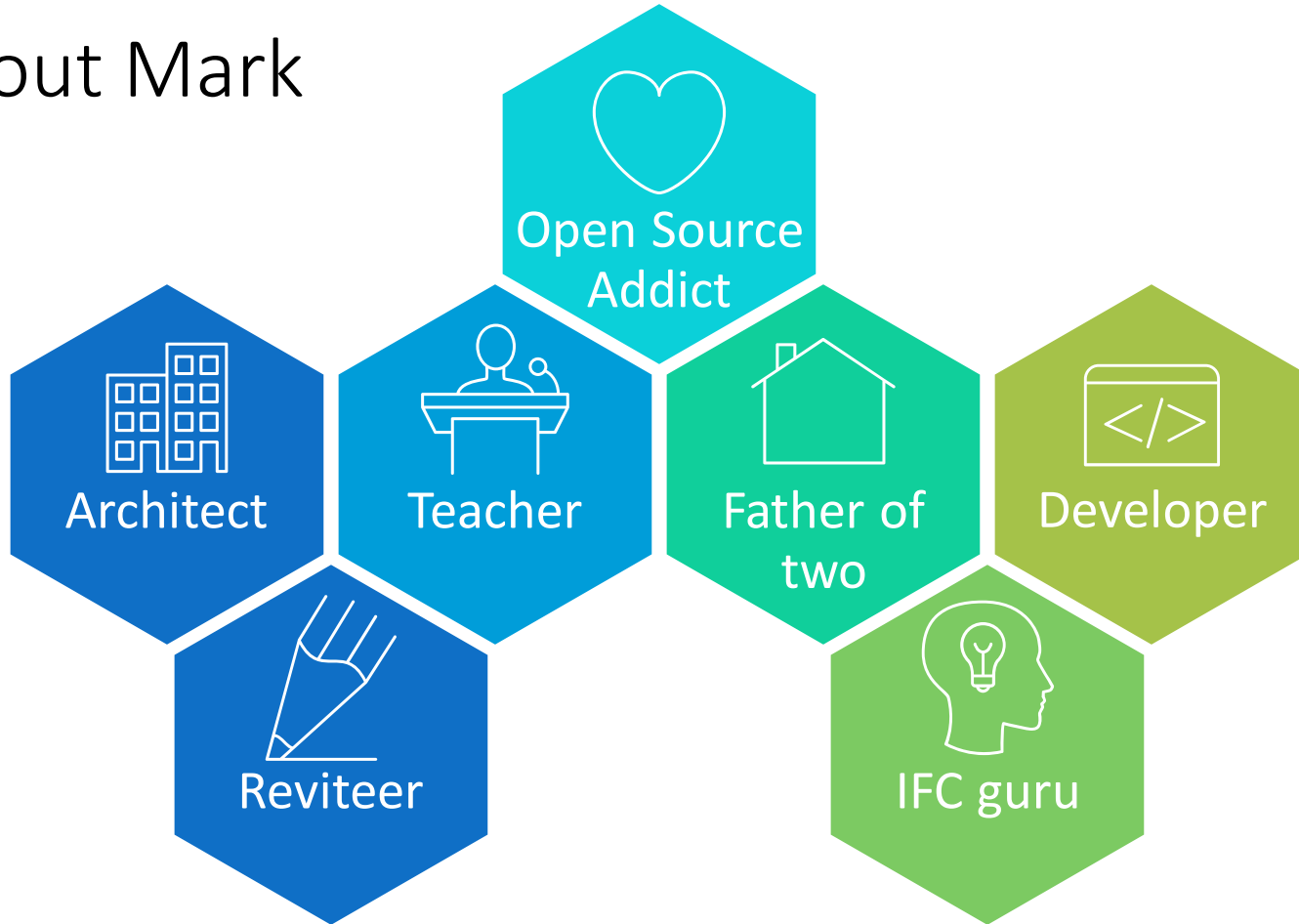


**Results & Future
development**

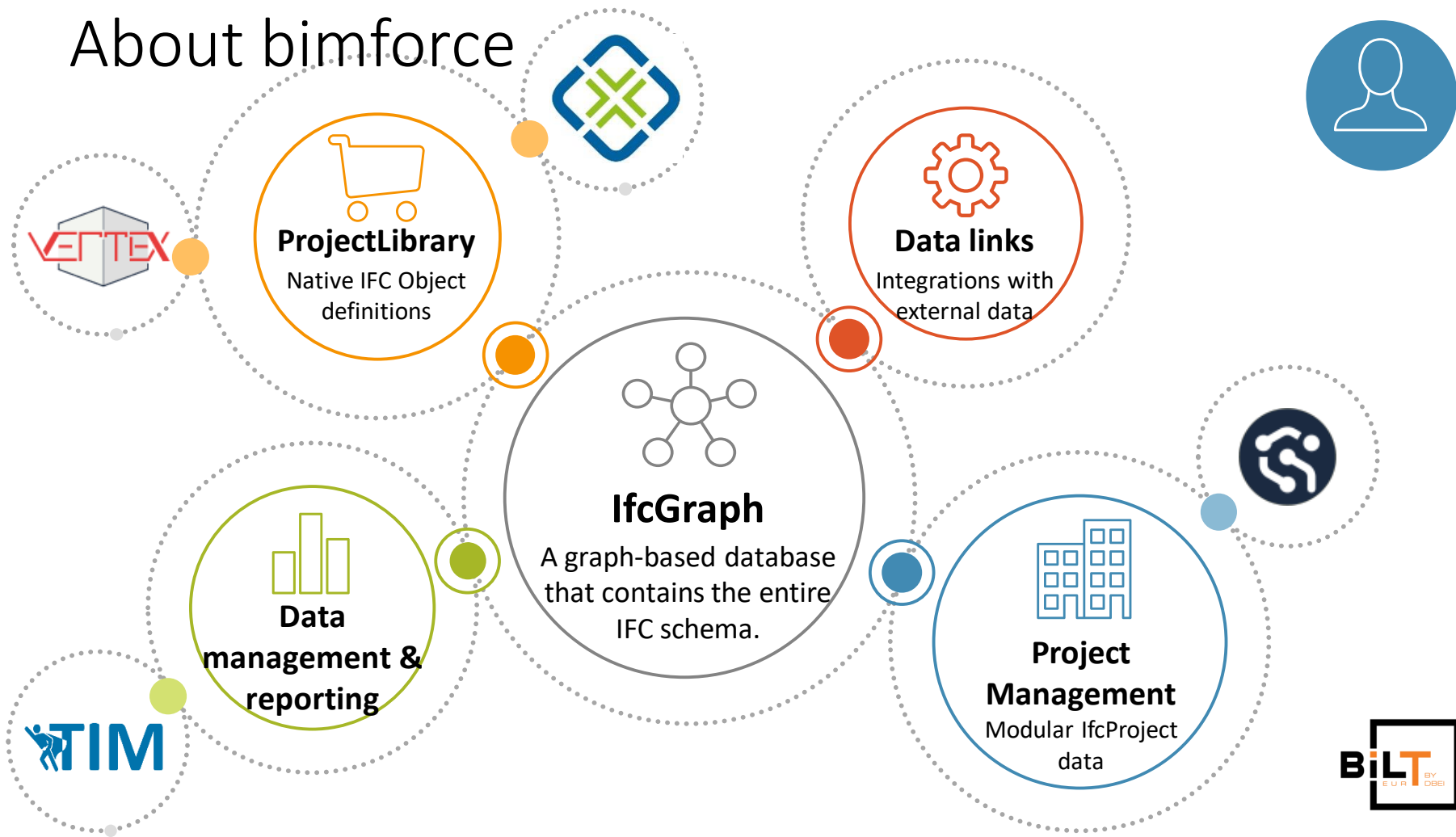
About Martijn



About Mark



About bimforce



About bimforce



GRiDS

- Common Information Environment
- Store all building data in a native IFC Graph database
- Connect with external datasources
- Bidirectional communication with external processes
- Visualisation of data in IFC viewers
- Add information to existing IFCs, extract partials, etc



Vertex

- Native IFC Project Library elements
- Support for full IFC or hybrid Revit solutions
- Support for Families as external geometry placeholder
- Material and Color libraries
- Online Content Management



Housing Configurator

- Fully customizable native IFC housing generator
- 21 housing types with a couple of thousand buyer options and unlimited material options.
- Potential buyers can customize their house with predefined options.
- Automatically creating drawings and contractual documents based on buyer preferences



TIM

- Technical Information Management at your fingertips.
- Define Classifications and add datasets to specific Classifications
- Use Graph Algorithms to automatically scan, review and update IFC models with predescribed information sets.

Client Question



Is it possible to develop a generator for single family houses that meets the following functional requirements:

1. No modelling software necessary to set up a project
2. GC can select appropriate options per building
3. GC can add contractual descriptions for options and attach a price tag

Client Question



Is it possible to develop a generator for single family houses that meets the following functional requirements:

4. Client can select desired options, generate a model, contractual drawings and an overview of picked options and their prices.
5. Data belongs to GC, and stays on their systems at all times.

Client Question



How it started...

12 building designs

139.968 possible configurations per type

(one hundred thirty-nine thousand nine hundred sixty-eight)

How it's going...

24 building designs

733.835.427.840 possible configurations per type

(seven hundred thirty-three billion eight hundred thirty-five million four hundred twenty-seven thousand eight hundred forty)

Configuring a project



Demonstrating the Housing Configurator:

- Setting up a project
- Setting up a site
- Client configuration

The screenshot shows the ModuFair Housing Configurator interface. At the top, there is a search bar with the text "Test vrijstaand" and a blue button with a plus sign. Below the search bar, there are buttons for "Show Archived", "Archive", "Nieuw", and "Toon als 3D". On the right side, there are buttons for "Project Creator", "Logout", and "Help". Below the search bar, there are buttons for "Importeer IFC" and "Verwijder IFC".

Project eigenschappen

ProjectNaam	
Project Omschrijving	
Startdatum	
Einddatum	
Bedrijfsnaam	
Naam projectleider (WoningConfigurator)	
Email projectleider (WoningConfigurator)	

Kavel eigenschappen

Kies kavelnummer	
Straat, huisnummer (toevoeging)	
Postcode, woonplaats	
Kadastraal adres	
Kaveloppervlakte	
Aantal gebouwen	
Woningtype	

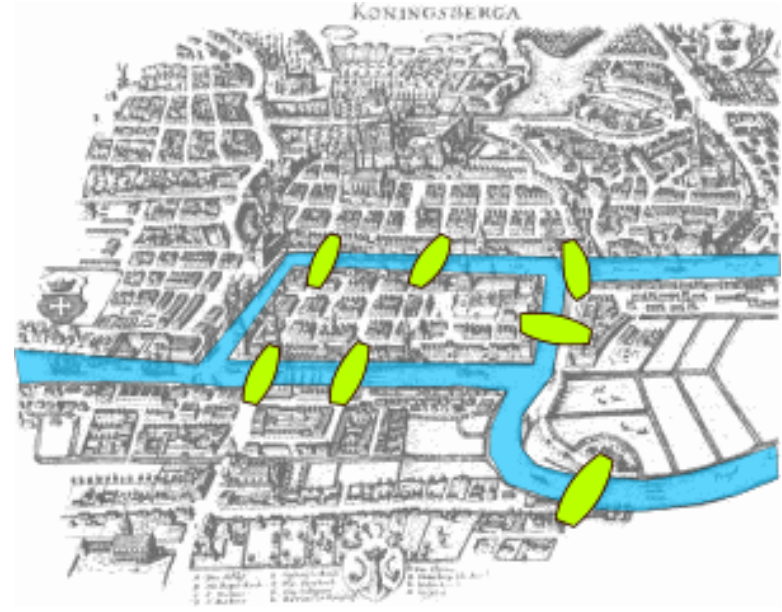
Toon		Link	Unlink
Straat	nr	toev.	
Postcode	Plaats		
Kadastraal adres			
1			
Kies woningtype			

IfcGraph



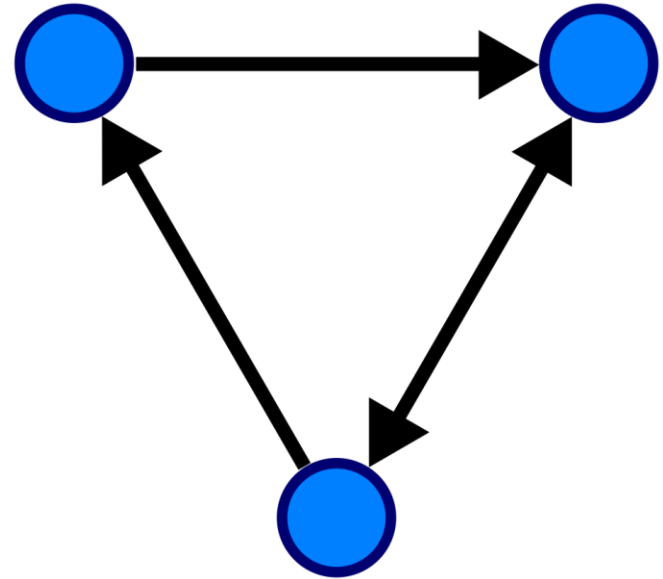
Graph theory is a mathematical theory, first described by Euler in 1736.

The problem he was solving is known as “The seven bridges of Konigsberg”



IfcGraph

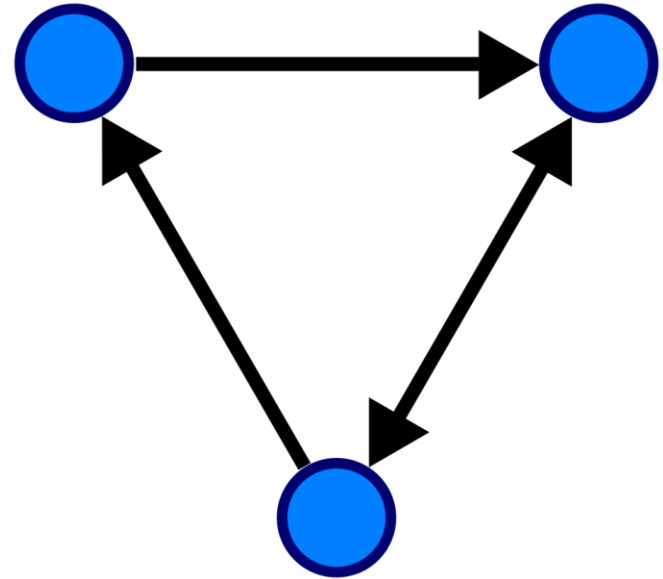
- A graph database differs from “traditional” databases in the way data is stored.
- A graph consists of vertices (nodes) and edges connecting them (relationships)
- Both nodes and relationships can have properties.
- Relationships can have a direction



IfcGraph

This makes Graphs perfectly suitable in situations where the importance of data is not in the data, but in the relationships between datapoints.

Which is exactly the case with IFC



IfcGraph



Why would you want to have a native IFC database?

1. Single definition / status. A file “is”. A database can have multiple “branches” simultaneously
2. Change management
3. Linked datasources
4. Parametrics

IfcGraph



But how do you get IFC in a graph?

A quick recipe:



IfcGraph



- Steep learning curve
- Manual coding of ifc datastructures
- Ifc issues
 - GlobalID missing for a lot of things
 - Missing concepts (mirror, array)
 - Unsorted lists
 - Annotations missing from IFC4

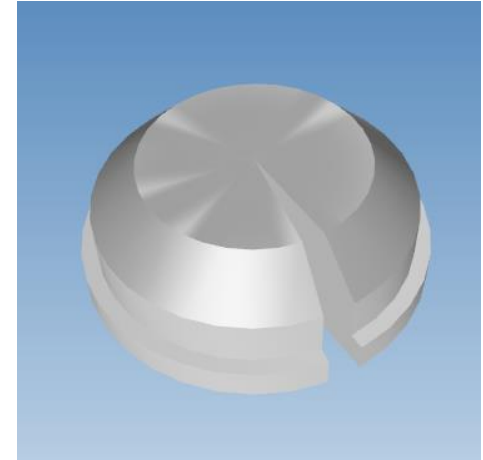
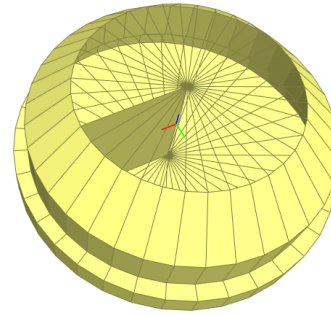
IfcGraph



- Implementation challenges:
 - Nobody uses IfcProjectLibrary
 - Aligning objects
 - Example: [Solar panels](#)
 - Grouping objects
 - Example: [Beam system](#)
 - Complex geometry: how do we NOT describe the complete geometry in IfcGraph
 - Example: [Stairs](#)
 - Reusing openings for multiple walls
 - Local Placement vs Global placement
 - Symbolic representation
 - Example: [Symbols](#)

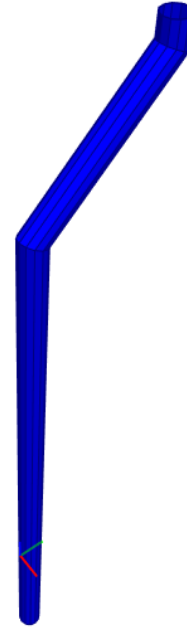
Trials and Tribulations

- Vast majority of geometry concepts not supported by 3rd party software.
- Including the ones that have most parametric capabilities
- For instance:
a simple revolve



Trials and Tribulations

- Circular shapes, bends

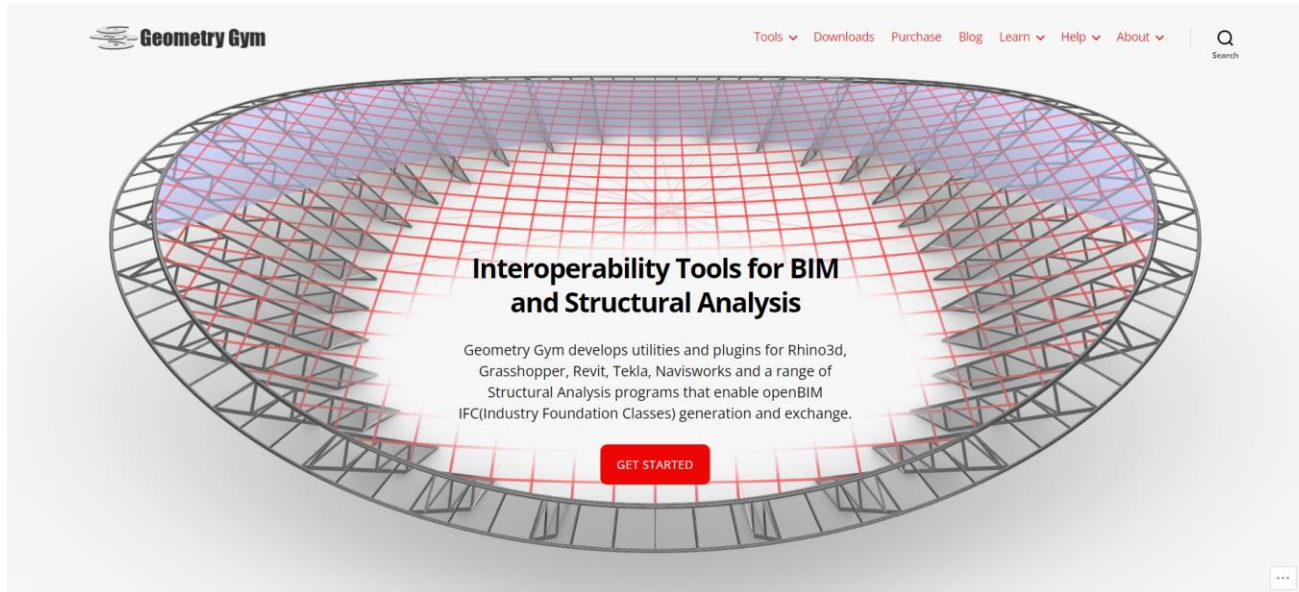


Trials and Tribulations



Parsing an IFC EXPRESS file to and from JSON.

MAJOR shoutout to the King of IFC, John Mirtschin



Trials and Tribulations

Testing 1-2-3(000)

Trials and Tribulations



- How do we view the results?
- How do we produce drawing sheets from IFC?



Developing an Open Source Viewer



MAJOR shoutout to Thomas Krijnen - IfcOpenShell

The screenshot shows the IfcOpenShell website homepage. At the top, there is a navigation bar with links for 'home', 'downloads', 'forums', 'blog', and 'documentation'. A funding progress bar indicates '\$504 / \$1,000 funded' with a green progress indicator. Social media icons for Twitter and GitHub are also present. The main content area features the 'IfcOpenShell' logo and tagline 'The open source IFC toolkit and geometry engine'. Below this, a paragraph describes the project's purpose: 'IfcOpenShell helps you develop digital platforms for the built environment. Read, write, and modify Building Information Models using IFC, a diverse digital language from design to construction and beyond.' A green button labeled 'Download v0.7.0' is provided. To the right, an illustration shows a computer monitor displaying a 3D building model, connected by lines to three separate 3D building models. Below the main content, there are four feature boxes: 1. 'A library you can rely on' - Developed by a community of hundreds of developers and trusted to deliver many AEC technologies that power our industry. 2. 'Any technology stack' - Build with Windows, Mac, or Linux, in C++ or Python. Deploy headless server tools, or author IFC natively with a rich graphical interface. 3. 'All schemas and formats' - Support IFC2X3, IFC4, IFC4.3, or load your own at run-time. Read and write IFC-SPF, IFCJSON, IFCXML, and IFCHEDF5. 4. 'Native IFC authoring API' - High level API for hundreds of tasks. Perform complex authoring like copying objects, cost calculation, or 4D simulation with one line of code. The bottom right corner of the screenshot shows a logo for 'BILT BY CUBI'.

home downloads forums blog documentation

\$504 / \$1,000 funded

IfcOpenShell
The open source IFC toolkit
and geometry engine

IfcOpenShell helps you develop digital platforms for the built environment.
Read, write, and modify Building Information Models using IFC,
a diverse digital language from design to construction and beyond.

Download v0.7.0

A library you can rely on
Developed by a community of
hundreds of developers and trusted to
deliver many AEC technologies that
power our industry.

Any technology stack
Build with Windows, Mac, or Linux, in
C++ or Python. Deploy headless
server tools, or author IFC natively
with a rich graphical interface.

All schemas and formats
Support IFC2X3, IFC4, IFC4.3, or load
your own at run-time. Read and write
IFC-SPF, IFCJSON, IFCXML, and
IFCHEDF5.

Native IFC authoring API
High level API for hundreds of tasks.
Perform complex authoring like
copying objects, cost calculation, or
4D simulation with one line of code.

BILT
BY
CUBI

Developing an Open Source Viewer

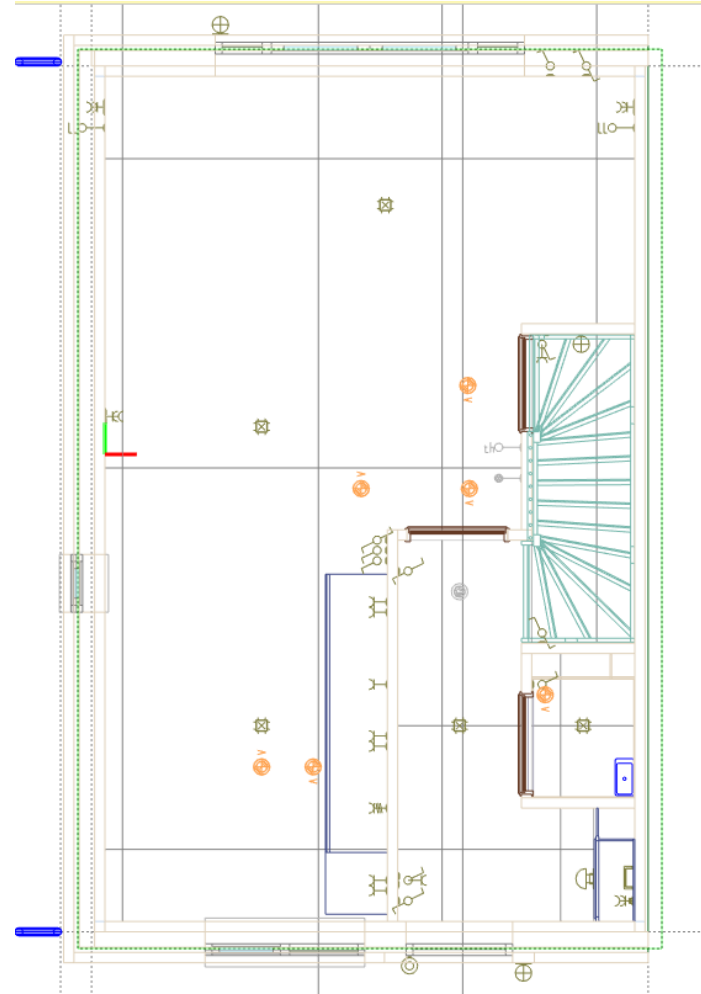


- Using Three.js as webbased viewer
- Well-known viewer (lots of potential developers & toolkits available)
- Perfectly suitable for smaller projects
- When we started (4yrs ago) not really used in AEC

Symbols & Dimensions

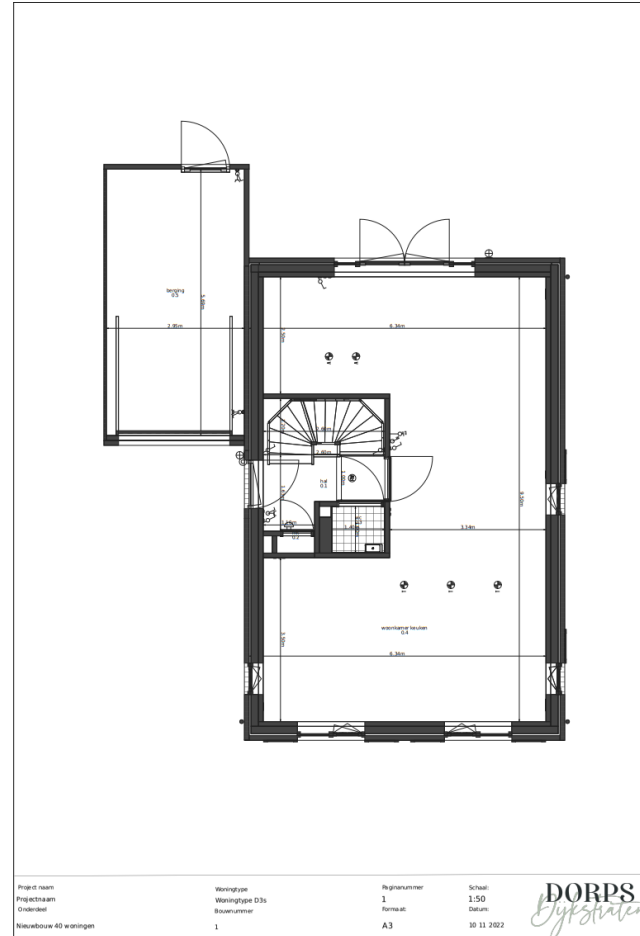
- Contrary to popular opinion, they do exist. Even in IFC4
- And, they are supported. At least, by some vendors.
- This is an example in “native” BIM Vision.
- Supports symbolic representations (in both 2D and 3D view).

Our own viewer does something extra



Symbols & Dimensions

- Based on Open Source initiatives
 - IfcOpenShell by Thomas Krijnen
 - BlenderBIM work by Dion Moul
 - Three.js visualisation platform
- Adapted the viewer to translate certain IfcAnnotations into text and dimensions.
- We can also define Views this way.
- And ceate pdf drawings.



Views



- Based on Open Source initiatives
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Building Blocks



A closer look at IfcProjectLibrary geometry

- What do we have?
- Support by 3rd party viewers
- How are we going to unlock this?

Configuring a project



Endresults



Future developments

- Speed, performance, stability and UI improvements
- Aggregating separate buildings in a single site plan
- Object configurator
- Improve import of IFC components in Revit
- Interactive object placement
- Assembly configurator
- Building configurator

One on One Meeting

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IFC Housing Configurator: Driving building automation with IFC

Mark Wieringa & Martijn de Riet

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